Kapasa Makasa University
Chinsali

SA 330 2025 Assignment 1

Instructions: Answer all questions, each question has 10 equal marks.

Question One

a). Explain the importance of keeping farm animals in Zambia as far as food security, income generation and industrial manufacturing of animal products are concerned?

b). What other types of industries in Zambia are associated with rearing of animals and how do these industries support the feeding and management of different animal species?

Question Two

a). What are the advantages and disadvantages of keeping ruminants compared to that of non-ruminants in Zambia?

b). What are the key limitations to increased animal production in the country and how can such limitations be addressed to promote and improve animal productivity?

Question Three

a). Explain the steps you may wish to take in order to provide enough feed for grazing animals under your care for them to have adequate supply of feed through out the year? Why are we only concerned with feeding of ruminants during the dry season?

b). What is proximate analysis and how are the different nutrients contained in animal feeds classified into different categories?

Question Four

a). Explain the importance of minerals and vitamins in animal feeding and how do animals meet requirements for these nutrients?

b). Explain the steps you may need to take in order to improve the productivity and utilization of natural grazing pastures?

Question Five

a). Explain the steps you may need to take for establishment and subsequent utilization of Rhodes grass (Chloris gayana) as an improved pasture and fodder crop in Zambia?

b). Explain the importance of planting tree and shrub legumes as fodder for grazing animals during the dry season?

Question Six

a). Explain the importance of legumes in animal feeding and what are some of the limitations associated with using such feeds to grazing animals?

b).Explain the steps you may wish to take for you to conserve grasses as hay including the nature and type of crops that you are likely to use?

Question Seven

a). Explain the steps you may wish to take for you to conserve grasses as silage including the nature and type of crops that you are likely to use? What precautions would you need to consider in order to come up with good silage?

b). Explain the steps you may wish to follow when selecting feed ingredients for use in formuating rations for non-ruminants under your care?

Question Eight

Using a Pearson Square, formulate a pre-lay ration to have 18% Crude Protein using Maize meal, Soya cake, Di-calcium Phosphate (DCP) and a vitamin pre-mix. The DCP and the premix are fixed at 2.0 and 0.7% of the ration, respectively? The nutrient composition of the required ingredients are as tabulated in Table 1. What are the energy, calcium and phosphorus contents in the formulated diet?

Table 1: Nutrient composition of feed ingredients used in the formulation of a layers pullet developer.

Ingredient	**Crude Protein %**	**Energy Kcal/kg**	**Calcium (%)**	**Phosphorous (g/kg)**
Maize	9.0	3400	0.04	0.3
Soya Cake	44.0	2950	0.40	0.65
DCP	0	0	24	18
Premix	0	0	0	0

Question Nine

Using algebraic expressions, formulate a broiler starter ration to have 22% crude Protein (CP) using Maize meal, Sorghum, Soybean cake, fish meal, Limestone, Di-calcium Phosphate (DCP) and a broiler premix. You are advised that the inclusion rates for lime, DCP and premix are fixed at 2.0, 1.0% and 0.5% of the ration, respectively. The Sorghum and Fish meal are also fixed at 18 and 4% of the ration, respectively. The concentration of nutrients in the required ingredients are presented in Table 2: How much energy, calcium and phosphorus is this ration expected to have?

Table 2: Nutrient composition of feedstuffs required for the formulation of the pig grower diet.

Ingredient	**Protein (%)**	**Digestible Energy***	**Calcium (%)**	**Phosphorus (%)**
Maize meal	9.0	3600	0.04	0.30
Sorghum	12.0	3700	0.03	0.30
Soy beans meal	45	2900	0.30	0.65
Fish meal	64	2800	1.8	3.2
Limestone	0	0	38	0
Di-calcium Phosphate	0.00	0.00	24.00	18.00
Pig grower Premix	0.00	0.00	0.00	0.00

- Mcal/kg

Question Ten

Using the deviation method, formulate a concentrate mixture to have 140 g/kg of digestible crude protein (dCP) using maize meal number 3, Sorghum, Sunflower cake, dicalcium phosphate (DCP) and a vitamin and mineral premix. You are advised that the DCP and the vitamin pre-mix are fixed 25 and 10 g/kg of the diet, respectively? If the nutrient composition of the ingredients is as given in Table 3, what are the energy, calcium and phosphorus contents of the concentrate mixture?

Table 3: The composition of nutrients (g/kg) in feed ingredients required for the formulation a protein feed concentrate.

Ingredients	**Digestible Protein**	**Total Digestible Energy**	**Calcium**	**Phosphorus**
Maize meal	75	800	4	3
Sorghum meal	85	720	3	3
Sunflower meal	230	580	3.3	10.8
DCP	0.0	0.0	240	180
Vitamin premix	0.0	0.0	0.0	0.0